DE–6149 **101**

DISTANCE EDUCATION

M.C.A.(NS) DEGREE EXAMINATION, MAY 2013.

DIGITAL COMPUTER ORGANIZATION

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Subtract 3250 from 72532 using l's and 2'scomplement. (10)

   (b) Convert the decimal number 625.625 to binary, octal and hexadecimal systems. (10)

2. (a) Reduce the following function by Karnaugh map and represent the reduced function in sum of products and product of sums from: (10)

   $F = \Sigma(1,5,6,7,14,15)$

   (b) Simplify the following Boolean functions. $F = (A, B, C, D) = \Sigma(0,1,5,8,9,10)$ using

   (i) Sum of products

   (ii) Products of sums

3. (a) Design a 3 bit binary counter and explain. (10)

   (b) Write notes on multiplexer. (10)

4. (a) Design a Half adder circuit. (10)

   (b) Explain the functions of shift registers. (10)

5. (a) Describe the linear select memory organization. (10)

   (b) Describe the needs for secondary memory. (10)

6. (a) Explain different types of ROMS in detail. (10)

   (b) List out the sequence of operations of control register and explain. (10)

7. (a) Discuss the magnetic tape & its working principles in detail. (10)

   (b) What is interrupts? Discuss about interrupts priorities in detail.

8. (a) Explain in detail about DMA. (10)

   (b) Discuss in detail about the I/O processor. (10)

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DE–6150 **102**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, MAY 2013.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Write brief notes on :

   (a) Word processing (10)

   (b) Integrated software and suites. (10)

2. Explain about video conferencing and picture phones. (20)

3. Describe about modems and data communication software. (20)

4. Explain in detail about cyberethics. (20)

5. Explain the following :

   (a) C characters (10)

   (b) Library functions in C. (10)

6. Explain the various control statements in C. (20)

7. Write brief notes on :

   (a) Multidimensional arrays (10)

   (b) Strings. (10)

8. Describe briefly about macros and inline functions. (20)

DE–6151 **103**

DISTANCE EDUCATION

M.C.A (N.S) DEGREE EXAMINATION, MAY 2013.

C AND DATA STRUCTURES

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write notes on structure and functions. (10)

   (b) What is a pointer? How do you create a file pointer? Give an example. (10)

2. (a) Write in detail about storage class initialization. (10)

   (b) Write a program in C to multiply two matrices of the order (3×3). (10)

3. (a) Explain briefly about the concept of stack. (10)

   (b) Write an algorithm for converting infix expression into postfix expression. (10)

4. (a) Define Queue. Explain the various operations of queue with example. (10)

   (b) Explain the various operations performed on doubly linked list. (10)

5. (a) Explain in detail about Huffman algorithms. (10)

   (b) What is a binary tree? Explain the linked representation of a binary tree. (10)

6. (a) Discuss briefly about the applications of liner linked list (10)

   (b) Explain in detail about binary tree traversal methods. (10)

7. (a) Discuss about the sequential search technique with suitable example. (10)

   (b) What is radix sort? Explain with algorithm and example. (10)

8. (a) Write notes on arrays in C. (10)

   (b) Explain Pointers to arrays. (10)

DE–6152 **104**

DISTANCE EDUCATION

M.C.A(NS). DEGREE EXAMINATION, MAY 2013.

OFFICE AUTOMATION

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write short notes on

   (i) Finding the document (5)

   (ii) Copying and moving the document. (5)

   (b) What is GUI? Discuss the uses of title bar, status bar, tool bar. (10)

2. (a) What are the different ways to create a word document? Explain in detail. (10)

   (b) Explain briefly about formatting the text in Ms-word. (10)

3. (a) Discuss in detail about printing and edition a document. (10)

   (b) Explain the following:

   (i) Spell checking. (5)

   (ii) Moving and resizing picture. (5)

4. (a) What is Worksheet? How does is differs from spreadsheet. (10)

   (b) Describe briefly about editing options available in worksheet. (10)

5. (a) Explain briefly about the steps to create a new database file in Ms-Access with example. (10)

   (b) Discuss on the following.

      (i) Adding, Editing and deleting records. (6)

      (ii) Hiding columns. (4)

6. (a) Describe briefly about report wizard in Ms-Access. (10)

   (b) What is primary key? What are the steps to create a primary key? (10)

7. (a) What are the various steps involved in creating a PowerPoint presentation? Explain in detail. (10)

   (b) Explain the procedure for adding animation effect and transition effect in PowerPoint presentation.(10)

8. Discuss on the following:

   (a) Creating a word handouts. (10)

   (b) Merging an access table with a word letter. (10)

DE–6153 **105**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

DISCRETE MATHEMATICS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Show that $P \to (Q \to P) \Leftrightarrow \neg P \to (P \to Q)$. (10)

   (b) Show that $\{\downarrow\}$ is a functionally complete set. (10)

2. (a) Obtain the *pdnf* and *pcnf* of the formula. (10)

   $(\neg P \to R) \wedge (Q \rightleftarrows P)$

   (b) Express $P \downarrow Q$ interms of $\uparrow$ only. (10)

3. (a) Show that $R \to S$ can be derived from the premises $P \to (Q \to S)$, $\neg R \vee P$ and Q. (10)

   (b) Show that from

   (i) $(\forall n)(H(n) \to m(n))$.

   (ii) $H(s)$ the conclusion $M(s)$ follows. (10)

4. (a) If R and S are relation from A to B show that $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$. (10)

   (b) Prove that the relation "congruence modulo *m*" over the set of positive integers is an equivalence relation. (10)

5. (a) Let $A = \{1,2,3,4\}$ and $R = \{(1,2),(2,3),(3,3),(3,4),(4,2)\}$ find the transitive closure of R. (10)

(b) Let $f : A \to B$ and $g : B \to C$ be both one-to-one and onto functions then prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. (10)

6. (a) Show that the set N of natural numbers is a semigroup under the operations $x * y = \text{man}\{x, y\}$. Is it a monoid. (10)

(b) Prove that the identity element of a group $(G,*)$ is unique. (10)

7. (a) Show that if every element in a group of its own inverse, then the group must be abelian. (10)

(b) Show that if $\langle G,* \rangle$ is a cyclic group, then every subgroup of $\langle G,* \rangle$ must be cyclic. (10)

8. (a) Draw the graph represented by the following adjacency matrix. (10)

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

(b) Represent the expression $((a-c)*d)/(a+(b-d)$ as a binary tree and write prefix form of the expression. (10)

DE–6154 **108**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

OBJECT ORIENTED PROGRAMMING AND C++

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain in detail about Class, Objects, Methods and Messages. (10)

   (b) Write about inheritance and polymorphism. (10)

2. (a) Discuss the various control structures in C++ with syntax and example program. (15)

   (b) Write about C++ standard I/O. (5)

3. (a) Explain the various file stream classes, file modes and file pointers. (15)

   (b) Define file I/O manipulators. (5)

4. (a) Write a C++ program to pass integer object as argument into another integer object and add the values. (12)

   (b) Define base class and derived class. (8)

5. (a) Write a C++ program to create a string class to contain character array. Declare the string objects. Overload the + operator to perform concatenation. (15)

   (b) Explain variable pointers and array pointers. (5)

6. (a) Write a program to overload a function named input( ) to initialize integer member or float member based on parameter passed to it. (12)

   (b) State the rules for overloading operators. (8)

7. (a) Define explicit and implicit type conversion. (6)

   (b) Write a C++ program to implement a type conversion operator. (14)

8. (a) Write a C++ program to implement virtual function and dynamic polymorphism. (10)

   (b) Explain reusing code with inheritance. (10)

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DE–6155 **109**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013

COMPUTER GRAPHICS

(2001 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Discuss the applications of computer graphics in briefly. (10)

   (b) Write short notes on Graphics system software. (10)

2. (a) What are the characteristics of good Line drawing algorithm. (10)

   (b) Write down the procedure for simple DDA Line drawing Algorithm. (10)

3. (a) Explain with an example, how to draw circle using Bresenham algorithm. (10)

   (b) Using examples, discuss the boundary fill algorithm. (10)

4. (a) What is 2D transformation? Explain with its matrix representation of the 2D transformation. (10)

   (b) Discuss briefly about 2D viewing transformations. (10)

5. (a) Explain in detail about Sutherland Hodgman Polygon Clipping Algorithm. (10)

   (b) Explain the different 3D viewing Schemes. (10)

6. (a) Write the Matrix representation for 3D composite transformations. (10)

   (b) Explain briefly about perspective projection. (10)

7. (a) Mention the limitation of Z – buffer algorithm clearly. (10)

   (b) Explain the practical significance of user interface and how the interface is designed. (10)

8. (a) What are the advantages of Interactive handling Models? Explain. (10)

   (b) Discuss on Keyword command Languages in briefly. (10)

DE–6156 **110**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

DESIGN AND ANALYSIS OF ALGORITHM

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. List out the basic steps involved in the complete development of an algorithm. (20)

2. (a) Define the terms network and adjacency vectors. Explain the algorithm connect with flowchart. (10)

   (b) What is Isomorphism? Explain adjacency powers matching technique with examples. (10)

3. Define linked lists. Write down the algorithm for insertion and deletion operations of linked list with flowcharts and examples. (20)

4. (a) Define the concept of working backward. Explain Jeep problem and find a solution for the same. (10)

   (b) Explain the definition of back tracking with an example of bicycle lock problem. (10)

5. (a) Define recursion. Explain how factorial and fibonacci series work using recursion. (10)

   (b) Write about doubly recursive function with necessary computations. (10)

6. Explain depth first search and breadth first search with detailed algorithms. (20)

7. List out the different sorting techniques. Explain their algorithms with flowcharts and sort the following data. 15, 10, 125, 30, 20, 35, 72, 85, 65, 90, 50. (20)

8. (a) Explain binary search algorithm with flow charts. Write down the steps involved to find data 20 from the given list 8, 20, 25, 30, 35, 45, 70, 73, 100, 125, 150. (10)

   (b) Discuss the algorithm for binary tree search and insertion with necessary sketches and examples. (10)

DE–6157 **111**

DISTANCE EDUCATION

M.C.A.(NS) DEGREE EXAMINATION, MAY 2013

ACCOUNTING AND FINANCIAL MANAGEMENT

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What are the golden rules of accounting?

   (b) What are the merits of Profit and Loss Account?

2. The following errors were found in the books of A.V.Thomas & Co. Give the necessary entries to correct them. (20)

   (a) Salary of Rs. 10,000 paid to Murali has been debited to his account.

   (b) Rs. 3,500 paid for a typewriter was charged to office expenses account.

   (c) Rs. 8,000 paid for furniture purchased has been charged to Purchase Account.

   (d) An amount of Rs. 5,000 withdrawn by the proprietor for his personal use has been debited to trade expenses account.

   (e) Rs. 2,000 received from Shanthi & Co has been wrongly entered as from Shakila & Co.

3. Prepare the Stores Ledger Account from the following information pricing the issues on FIFO method. (20)

Receipt of Material

1st January 2011,	300	tons	@ Rs. .8 per tonne
7th January 2011,	200	tons	@ Rs. .8.20 per ton
15th January 2011,	250	tons	@ Rs. .7.90 per ton
23rd. January 2011,	400	tons	@ Rs. .7.50 per ton
28th January 2011,	200	tons	@ Rs. .7.70 per ton

Issue of Material

5th January 2011,	140	tons	8th January 2011, 60 tone
15th January 2011,	120	tons	18th January 2011, 210 tons
22nd January 2011,	200	tons	25th January 2011, 150 tons
29th January 2011,	250	tons	

4. From the following data given by the Personnel department calculate Labour Turn over rate by applying

(a) Seperation Method (20)

(b) Replacement Method

(c) Flux Method

Number of workers on the Payroll:

At the beginning of the month 900

At the end of the month 1100

During the month 10 workers left, 40 persons were discharged. and 150 workers were recruited. Of these, 25 workers are recruited in the vacancies of those leaving, while the rest were engaged for an expansion scheme.

5. Balance Sheets of Big B' & Co Ltd as on 1.1.2008 and 31.12.2008 were as follows: (20)

Balance Sheet

Liabilities	1.1.2008	31.12.2008	Assets	1.1.2008	31.12.2008
Creditors	40,000	44,000	Cash	10,000	7,000
Mr B's Loan	25,000		Debtors	30,000	50,000
Loan from SBI	40,000	50,000	Stock	35 000	25,000
Capital	1,25,000	1,53,000	Machinery	80,000	55,000
			Land	40,000	50,000
			Building	35,000	60,000
	2,30,000	2,47,000		2,30,000	2,47,000

During the year machine costing Rs. 10,000 (accumulated depreciation Rs. 3,000) was sold for Rs. 5,000. The provision for depreciation against machinery as on 1.1.2008 was Rs. 25,000 and on 31.12.2008 Rs. 40,000. Net profit for the year 2008 amounted to Rs. 45,000. You are required to prepare the Funds flow Statement.

6. Fromthe following figures, you are required to calculate (20)

(a) P/V Ratio

(b) Break Even Point

(c) Margin of Safety

(d) Profit.

Sales Rs. 4000 Variable cost Rs. 2,000 Fixed Cost Rs. 1,600.

7. BB & Co, Chennai has prepared the following budget estimates for the year 2009—10. (20)

Sales units 15,000

Fixed expenses Rs. 34,000

Sales Value Rs. 1,50,000

Variable Cost Rs.6 per unit.

Calculate the P/v Ratio, break even point and margin of safety in each of the following cases.

(a) Decrease of 10% in selling price

(b) Decrease of 10% variable cost

(c) Increase of sales volume by 2000 unites

(d) Increase of— Rs. 6,000 in fixed costs.

8. From the following information given below prepare Flexible budget for 60 and 80 per cent capacities. (20)

	At 60% capacity Rs.	At 75% capacity Rs.	At 80% capacity Rs.
Variable Over head			
Indirect Material		7,500	
Indirect Labour		22,500	
Semi—Variable overheads			
Electricity (40% Fixed 60% Variable)		37,500	
Repairs and Maintenance (80% Fixed 20% Variable)		3,750	
Fixed overheads:			
Salaries		1,00,000	
Insurance		5,000	
Depreciation		25,000	

DE–6158 **112**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, MAY 2013.

COMMUNICATION SKILLS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. Briefly explain the effective communication.
2. Narrate different communicative skills.
3. Explain reading techniques.
4. Describe the Art of listening and the benefits derived from it.
5. What are the etiquettes of telephonic conversation? Note down the does and don'ts of a telephonic conversation.
6. Explain briefly the process of group discussion.
7. What are the tools for the personality identification?
8. Write short notes on the following: (4 × 5 = 20)
   (a) Body language
   (b) Report Writing
   (c) Managing process of negotiation
   (d) Inter personal communication.

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DE–6159 **201**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

RESOURCE MANAGEMENT TECHNIQUES

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Define redundant constraint. (4)

(b) Solve the LPP (16)

Maximize $Z = 2x_1 + x_2 + x_3$

Subject to

$4x_1 + 6x_2 + 3x_3 \leq 8$

$3x_1 - 6x_2 - 4x_3 \leq 1$

$2x_1 + 3x_2 - 5x_3 \geq 4$

$x_1, x_2, x_3 \geq 0.$

2. (a) What is meant by degeneracy of a LPP? (4)

(b) Using dual simplex method to solve the LPP (16)

Maximize $Z = -3x_1 - 2x_2$

Subject to

$x_1 + x_2 \geq 1$

$x_1 + x_2 \leq 7$

$x_1 + 2x_2 \geq 10$

$x_2 \leq 3$

and $x_1, x_2 \geq 0.$

3. (a) What is unbalanced transportation problem? How to solve it? (4)

(b) Solve the following transportation problem. (16)

	I	II	III	IV	Supply
F_1	14	56	48	27	70
F_2	82	35	21	81	47
F_3	99	31	71	63	93
Demand	70	35	45	60	

4. (a) Explain how maximization problems are solved using assignment model technique? (4)

(b) Solve the following assignment problem to find minimum cost (16)

Machines

		M_1	M_2	M_3	M_4
	J_1	5	7	11	6
Jobs	J_2	8	5	9	6
	J_3	4	7	10	7
	J_4	10	4	8	3

5. (a) What are the basic characteristic of a queue system? (4)

(b) Cost arrive at a petrol pum p, having one petrol unit, in poisson fashion with an average of 10 cars per hour. The service time is distributed exponentially with a mean of 3 minutes. Find (i) Average number of cars in the system (ii) Average waiting time in the queue (iii) Average queue length (iv) the probability that the number of cars in the system is 2. (16)

6. (a) Define Buffer stock. (4)

(b) A manufacturer has to supply his customer with 600 units of his products per year shortage are not allowed and storage cost amounts to Rs. 6 per unit per year. The set of cost is Rs. 800 find

(i) The economic order quantity

(ii) The minimum average yearly cost

(iii) The optimum number of orders per year

(iv) The optimum period of supply per optimum order. (16)

7. (a) What is resource levelling is network analysis? (4)

(b) Draw the network and determine the critical path also find various float of each activity. (16)

Activity:	1-2	1-3	2-4	2-5	3-4	4-5
Duration:	8	4	10	2	5	3

8. (a) What do you understand by group replacement? (4)

(b) The cost of a machine is Rs. 6,100 and its scrap value is Rs. 100. The maintenance costs found from experience are as follows. (16)

Year:	1	2	3	4	5	6	7	8
Main cost	100	250	400	600	900	1200	1600	2000

When should the machine be replaced?

DE–6160 **202**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

Non Semester

OPERATING SYSTEMS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Discuss about the structure of Operating Systems with neat diagram. (10)

   (b) What are system calls? How can you make a system call? — Explain. (10)

2. Write in detail about Inter Process Communication. (20)

3. Explain the various process scheduling algorithms with suitable example. (20)

4. Write a note on the following

   (a) Device controllers (10)

   (b) Interrupt handlers (10)

5. What is dead lock? How can you deal with deadlock? -Explain in detail. (20)

6. (a) Give a brief description about multiprogramming without swapping and paging. (10)

   (b) Write short notes on following

      (i) List method (5)

      (ii) Buddy systems. (5)

7. Discuss the usage of segments in memory management in detail. (20)

8. (a) Explain about File system reliability. (10)

   (b) Describe about the performance of file systems. (10)

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DE–6161 **203**

DISTANCE EDUCATION

M.C.A.(NS) DEGREE EXAMINATION, MAY 2013.

RDBMS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. (a) Explain the different user interfaces. (10)

   (b) Describe the advantages of database systems. (10)

2. (a) Give an account on Relational calculus with example. (10)

   (b) Discuss on Data manipulation and Data definition facilities with examples. (10)

3. (a) Explain Relational Database properties. (10)

   (b) Describe the Oracle database Architecture. (10)

4. (a) Explain DML commands with example. (10)

   (b) With suitable example, discuss Unions and Multiple part Queries. (10)

5. Example the following with example :

   (a) Creating Oracle Table. (10)

   (b) Table Definition. (10)

6. (a) Create a database for bus reservation system. Convert your database into 3NF. (10)

   (b) List down the advantage of representation data in the normalized form. (10)

7. (a) Discuss on major categories of SQL statements. (10)

   (b) Explain various comments on the Relational model. (10)

8. (a) Describe the language supported by Oracle precompiler using Embedded SQL. (15)

   (b) Write about Embedded SQL statement. (5)

DE–6162 **204**

DISTANCE EDUCATION

M.C.A (NS) DEGREE EXAMINATION, MAY 2013.

UNIX AND SHELL PROGRAMMING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss briefly about shell programming. (8)

   (b) Write short notes on the awk pattern scanning. (7)

   (c) Briefly explain about Directory hierarchy. (5)

2. (a) Explain about command line structure in shell. (7)

   (b) Discuss about shell variables in detail. (8)

   (c) How to use evaluating arguments in shell? (5)

3. (a) Write down the procedure for pick command in detail. (7)

   (b) Write short notes on Traps. (6)

   (c) Briefly explain about News command with suitable example. (7)

4. (a) Explain in detail about signal and Interrupts in Unix System. (10)

   (b) Discuss about Directories and inodes in Unix System. (10)

5. Explain in detail about the features in Unix. (20)

6. (a) Discuss about the various relational operators with suitable examples. (10)

   (b) Discuss on Four function calculator in detail. (10)

7. (a) Explain in detail about file access in Unix. (10)

   (b) Briefly explain about the features of accessing the environment in Unix. (10)

8. Explain the following:

   (a) Troff level (5)

   (b) Tbl and eqn preprocessors (5)

   (c) Error recovery (5)

   (d) Maual page. (5)

DE–6163 **205**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

COMPUTER NETWORKS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about network hardware. (5)

   (b) Discuss about OSI reference model. (15)

2. (a) Give a brief note on transmission media. (10)

   (b) Write about cellular radio. (10)

3. (a) Explain about error detection and correction codes. (10)

   (b) Give a brief note on sliding window protocol. (10)

4. (a) Explain about congestion control algorithms. (10)

   (b) Write notes on :

      (i) Fragmentation (5)

      (ii) IPV6 (5)

5. (a) Explain about ICMP protocol. (10)

   (b) Give a brief note on UDP. (10)

6. (a) Explain the problems associated with flow control and buffering. (10)

   (b) How will you measure the performance of networks? Explain. (10)

7. (a) Give brief note on public key algorithms. (10)

   (b) Explain SNMP in detail. (10)

8. (a) Discuss Electronic Mail in detail. (10)

   (b) Write notes on WWW. (10)

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DE–6164 **208**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

SIMULATION AND MODELLING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Differentiate between static and dynamic mathematical models. (10)

   (b) Write in detail the numerical computational technique for continuous models. (10)

2. (a) List out and explain the principles used in modeling. (10)

   (b) Write notes on feedback systems. (10)

3. (a) Explain exponential growth models with suitable examples. (10)

   (b) What are multi-segment models? Explain in detail. (10)

4. (a) With a suitable diagram, explain the structure of a system dynamics model. (10)

   (b) Explain how coding is carried out in DYNAMO language for dynamic models. (10)

5. (a) Differentiate between discrete and continuous probability functions. (10)

   (b) Discuss in detail the computer generation of random numbers. (10)

6. (a) Write notes on arrival patterns. (10)

   (b) List out and explain the measures of queues. (10)

7. (a) Explain in detail the delayed calls. (10)

   (b) Explain how distributions and transit times are recorded. (10)

8. (a) Discuss in detail the GPSS block types. (10)

   (b) Differentiate facilities and storages in system simulation. (10)

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DE–6165 **209**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

OBJECT ORIENTED ANALYSIS AND DESIGN

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the advantages of object oriented approach. (10)

   (b) Briefly explain the elements of object model. (10)

2. (a) What is an object? Explain its nature. (10)

   (b) Describe various kinds of relationships among classes. (10)

3. (a) Explain the various object oriented analysis approaches. (10)

   (b) Write short notes on key abstraction mechanism. (10)

4. (a) Explain the relationships that are possible among the classes in the UML representation with your own example. (10)

   (b) What do you mean by deployment diagrams? Explain them with a suitable example. (10)

5. (a) Draw the class diagram and usecase diagram, explain. (10)

(b) Describe the basic activities object oriented analysis and explain how usecase modeling is useful in analysis. (10)

6. (a) Explain the following Sequence diagram interaction diagrams. (10)

(b) Explain the following Collaboration diagram interaction diagrams. (10)

7. (a) What are patterns? (5)

(b) Explain how classes are identified using common class patterns. (15)

8. (a) Explain Package diagram with example. (10)

(b) Explain Interaction diagram with example. (10)

DE–6166 **210**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

INTERNET PROGRAMMING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the overview of Internet Programming. (10)

   (b) Describe briefly the WWW Design issues. (10)

2. (a) Give a brief account on Perl in detail. (10)

   (b) Explain the features of CGI. (10)

3. Give a detailed note on Java script and its components. (20)

4. (a) Explain the various Internet Markup Languages in detail. (10)

   (b) Write short notes on Microsoft Internet Explorer. (10)

5. (a) Write short notes on Active X controls. (10)

   (b) Discuss the process of creating Netscape Navigator plug-Ins. (10)

6. (a) Explain pulling web information. (10)

   (b) Write an essay about multiple protocols. (10)

7. (a) Explain the function done on Images and Hyperlinks. (10)

   (b) Write the steps for framing your graphics. (10)

8. (a) Explain about site and page design. (10)

   (b) Bring out the significance of easy web graphics. (10)

DE–6167 **211**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

SOFTWARE ENGINEERING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. Explain the following:

   (a) Linear sequential model. (10)

   (b) Prototyping model. (10)

2. (a) Discuss briefly about Project Estimation. (10)

   (b) Describe various Emperical Estimation Models. (10)

3. Write brief notes on

   (a) Software Quality assurance. (10)

   (b) Software reviews. (10)

4. Explain the following

   (a) Analysis concepts and principles. (10)

   (b) Software prototyping. (10)

5. (a) Explain briefly about ISO 9000 Quality standards. (10)

   (b) Discuss about Risk Mitigation. (10)

6. (a) What is Behavioural Modeling? Explain. (10)

   (b) List the various design concepts and explain. (10)

7. Explain the various Design Methods. (20)

8. Write short notes on

   (a) White box Testing. (10)

   (b) Black box Testing. (10)

DE–6168 **212**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

VISUAL PROGRAMMING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain and compare the different programming paradigms. (10)

   (b) Discuss about derived data types and Hungarian notation in windows programming. (10)

2. (a) Describe the methods used for obtaining device context handle. (10)

   (b) With an example explain the steps to create user defined DLL. (10)

3. (a) What are the different types of modules in VB. Explain? (10)

   (b) Explain the various looping structures available in VB. (10)

4. (a) Discuss the different types of procedure that can be created using VB. (10)

   (b) How will you create a menu? Write the procedure to add menu commands. (10)

5. (a) Write a program to insert, update and retrieve employee name and number from a database. (10)

   (b) Write the steps involved in the process of dialog box creation. (10)

6. (a) Write short notes on Recordset objects. (10)

   (b) With a neat sketch, describe the document view architecture in VC++. (10)

7. (a) Explain how MFL windows application uses the various resources in VC++. (10)

   (b) Write about splitter windows and its types. (10)

8. (a) Explain the concept of object linking and embedding with sample VC++ application. (10)

   (b) What is ODBC? Describe its architecture. (10)

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DE–6169 **301**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

SOFTWARE PROJECT MANAGEMENT

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Discuss in detail about any three software process model with neat sketch. (20)

2. Write a brief note on critical path method (CPM) with an appropriate example. (20)

3. Explain the project tracking meetings and their outcomes in detail. (20)

4. (a) Describe the importance of requirements gathering. (10)

   (b) What do you mean by testing? Discuss any two types of testing strategies with neat sketch. (10)

5. (a) How to maintain vendor relationship? Explain with appropriate examples. (10)

   (b) What is quality system? Discuss the essential parameters used for measuring the quality of a product. (10)

6. Explain briefly about the steps involved in tailoring and improving the process. (20)

7. Discuss in detail about the various approaches for identifying the activities. (20)

8. (a) Write a brief note on post project reviews. (10)

   (b) Write short notes on the following :

      (i) Sub contractors. (5)

      (ii) Performance incentives. (5)

DE–6170 **302**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

CLIENT SERVER TECHNOLOGY

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What do you mean by client server? Explain its important benefits. (10)

   (b) Explain the various ways to improve the performance of client server. (10)

2. Write a brief note on UNIX operating system. (20)

3. Explain client/server development methodology and discuss the various phases and their activities involved in system integration life cycle. (20)

4. What do you mean by DCOM? Explain the architecture and its components with neat sketch. (20)

5. Describe the following :

   (a) SQL windows. (10)

   (b) Power builder. (10)

6. What do you mean by network management? Discuss briefly about the five OSI management areas with suitable illustrations. (20)

7. Write short notes on the following :

   (a) Token ring. (5)

   (b) Network file system. (5)

   (c) Distributed computing environment. (10)

8. Describe the server functionality in detail. (20)

DE–6171 **303**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

PATTERN RECOGNITION AND IMAGE PROCESSING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the Image acquisition process in detail. (10)

   (b) Explain about satellite images. (10)

2. (a) Explain the Raster screen in detail. (10)

   (b) Discuss about the Image processors. (10)

3. (a) Explain the crack edge detection with an example. (10)

   (b) Explain the Morphological operation in detail. (10)

4. (a) Explain the quantizing Compression method. (10)

   (b) Explain the jpeg standard in detail. (10)

5. (a) Explain how to measure the similiarity clustering criteria. (10)

   (b) Explain the advantages of pattern classification techniques. (10)

6. (a) Explain the Maximum distance pattern classification. (10)

   (b) What are the characteristics of likelihood function? Explain. (10)

7. (a) Discuss about the Isodata algorithm in detail. (10)

   (b) Discuss the features of pattern recognition system. (10)

8. (a) Explain with examples the syntactic pattern recognition. (10)

   (b) Discuss about syntactic recognition learning process in detail. (10)

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DE–6172 **304**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

COMPILER DESIGN

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) List out the different compiler writing tools and explain them. (10)

   (b) Explain the role of lexical analyzer and the concept of input buffering. (10)

2. Write short notes on the following :

   (a) The notational conventions used to represent context free grammars. (10)

   (b) Concept of ambiguity and the method to eliminate ambiguity. (10)

3. Explain the algorithm of operator precedence parsing. Use the grammar.

   $E \to E + E$

   $E \to E * E$

   $E \to E \uparrow E$

   $E \to (E)$

   $E \to id.$

   Parse the string $id_1 * (id_2 \uparrow id_3)$. (20)

4. Find the canonical collection of LR(O) items for the following grammar : (20)

$S \rightarrow L = R$

$S \rightarrow R$

$L \rightarrow *R$

$L \rightarrow id$

$R \rightarrow L.$

5. (a) Explain the steps involved in syntax directed translation for arithmetic expressions. (10)

   (b) Compare the different notations used for representing syntax directed translations. (10)

6. Briefly explain the following :

   (a) Loop optimization (10)

   (b) DAG representation of basic blocks. (10)

7. Write the steps involved in the implementation of simple code generator with suitable example. (20)

8. Enumerate all possible errors in the different phases of compiler and briefly explain the technique used to recover from those errors. (20)

DE–6173 **305**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

MULTIMEDIA SYSTEMS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain in detail about the need and benefits of multimedia. (10)

   (b) Briefly describe about the multimedia platforms. (10)

2. (a) Discuss about the cross platform compatibility. (10)

   (b) Explain about the commercial tools and standards. (10)

3. (a) Discuss in detail about the non temporal media. (10)

   (b) Explain about temporal media. (10)

4. (a) Describe about extended images. (10)

   (b) Discuss in detail about the digital link. (10)

5. (a) Briefly explain about the Digital video. (10)

   (b) Explain about the video compression techniques. (10)

6. (a) Describe about the image compression standards. (10)

   (b) Explain about the classes and related items in object oriented multimedia. (10)

7. (a) Explain about the transform classes and format classes. (10)

   (b) Briefly explain about the CD–family and media types. (10)

8. (a) Explain in detail about Multimedia Architecture and operations. (10)

   (b) Discuss about the Multimedia and single user. (10)

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DE–6627 **106**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

LAB I — DATA STRUCTURES USING "C" AND MS OFFICE

(2003 and 2004 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Examiner has to select TWO questions choosing ONE from Data.

Structures and another from MS-Office.

DATA STRUCTURES USING 'C'

1. Write a C program to find the roots of a quadratic equation using function definition.

———————————— cut here ————————————

2. Write a C program to read 10 values to an array variable. Use pointers to locate and display each value.

———————————— cut here ————————————

3. Write a C program to implement insert and delete operations on queue using array concepts.

———————————— cut here ————————————

4. Write a C program to implement insert and delete operations on Linked list structure.

———————————— cut here ————————————

5. Write a menu driven program in C to find an element using Linear and Binary search methods.

MS - OFFICE

6. Type the 4 columns and five rows values in a document and convert into table format with headings.

---------- cut here ----------

7. Prepare a document to advertise your company and tender notice using newspaper format.

---------- cut here ----------

8. Create Worksheet in MS-EXCEL with the following details :

   Regno., Name, Mark 1, Mark 2, Mark 3, Total, Average

   (a) Using formula and function find the Total, Average, Maximum, Minimum total marks.

   (b) Sort the names in alphabetical order.

   (c) Create the bar chart for average mark with proper titles, legend and Gridlines.

---------- cut here ----------

9. An invoice table has the following details,
   Name of the item, Price of the each item, Number of quantity, item purchase, discount rate.

   (a) Type the data in the above format in MS-Excel.

   (b) Find the Total price, Average using built-in functions.

   (c) Sort the name in alphabetical order.

   (d) Draw a Pie chart for total price.

10. Create shop details table with the following fields in MS-Access :

Field Name	Data Type
Pcode	Text
Pname	Text
Quantity	Numeric
Price	Numeric

Execute the following queries using the shop details table :

(a) Select the quantity whose price greater than 1000.

(b) Select product whose product code is HDD555.

(c) Design the form to enter the data in the table.

DE–6628 **113**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

LAB II — GRAPHICS USING C++ AND ALGORITHMS

(2003 and 2004 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

TWO questions are to be selected by the examiner and the candidate should select the question by lot system.

1. Write a C++ program to draw nonlinear object ellipse.

---------- Cut here ----------

2. Write a C++ program to show graphical transformation on three dimensional objects.

---------- Cut here ----------

3. Draw a ball. Perform animation on it.

---------- Cut here ----------

4. Write a C ++ program to draw a straight line using simple DDA technique.

---------- Cut here ----------

5. Write a program for quick sort.

6. Develop a program for JEEP problem using working backward method.

---------- Cut here ----------

7. Write a program for 8-Puzzle problem using backtracking.

---------- Cut here ----------

8. Write a program for recursion. (Factorials and Fibonacci).

DE–6629 **206**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

LAB III — RDBMS AND SHELL PROGRAMMING

(2003 & 2004 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Each candidate should select one question using lot system.

Each subdivision carries 50 marks.

Break up of marks:

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. (a) A daily sales file contain record with the following fields : department number, date, item description, sales price for each item, quantity and cost of each item. Write a program using oracle to list all the input data and compute total amount of sales and profit for each sale. Accumulate and print total of sales and profit. The output report contains department number, item description, sales price, quantity, cost and profit.

   (b) Write a menu driven shell program to perform the following tasks.

   (i) Enter the sentence in file

   (ii) Search a given whole word in an existing file

   (iii) quit.

2. (a) A hospital maintains blood donors record information in a file. The items are donor number, donor name, donor age, donor address, pin, place of birth, blood group (A, B, AB and O). Write a program to print out the number, name and address of the donors for the following categories.

   (i) blood donor having the blood group AB

   (ii) blood donor in the age group between 16 and 25

   (iii) female donors having blood group A in the age between 20 and 25.

   (b) Write a menu driven shell program to copy, edit, rename, delete a file.

---------- Cut here ----------

3. (a) Create a table having the structure.

   Code for domestic and industry

   Current rates reading

   Previous rates reading.

   (i) Write a program to prepare the report in the following order:

Code	P.R	C.R	Amount
–	–	–	–

   (ii) List out the code and amount which are more than 100 units according to the code wise.

   (b) Write a shell program which accepts the name of a file from the standard input and then perform the following tests on it :

   (i) Enter the 5 names in a file

   (ii) Sort the names in existing file

   (iii) List unsorted and sorted file

   (iv) quit.

4. (a) Each morning a news paper vendor buy newspapers in wholesale from a distributor for 60 paise. He sell them in retail for 75 paise. At the end of the day the unsold papers are returned to the distributor for a 30 paise rebate per paper. Write oracle program to prepare a report for the newspaper vendor in the following format with 10 weeks data :

Week	Bought	Sold	Return	Profit	Loss
1	50	30	XX	XX.XX	XXX.XX
2	75	62	XX	XX.XX	XXX.XX
Total					

(b) Write a menu driven shell program for the following:

(i) List of files (ii) processes of users

(iii) today's date (iv) users of system

(v) quit to Unix.

---------- Cut here ----------

5. (a) A salary statement contains name, basic pay, allow, total deduction (includes IT), Gpay, Npay.

Allowance = 20% of basic pay;

Gross pay = Basic pay + Allowance

Deduction = 10% of basic pay

It is calculated on the basis of annual income under the following condition.

Annual salary	Income tax amount
<= 30,000	Nil
>30,000 but <55,000	30% of the excess over the amount Rs. 30,0000
<= 55,000	50% of the excess over the amount Rs. 55,000

Total deduction = Deduction + Income tax.

Write a program in SQL to prepare salary report for 5 employees.

(b) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display "pattern found", else display "error message". Also check if right amount of arguments are entered.

6. (a) An examination has been conducted to a class of 7 students and four scores of each student have been provided in the data along with register number, name. Write SQL program to do the following :

Assign a letter grade to each student based on the average score and list out the students register number and average score, grade. The minimum pass for each student is 50.

The grading system :

Average score	Grade
90–100	A
75–89	B
60–74	C
50–59	D
Less than 49	F-Fail

(b) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

(i) File existence

(ii) File readable

(iii) File writable

(iv) Both readable and writable.

7. (a) Write a program in SQL to hospital billing system having the following fields : Patient No, Patient name, age, doctor attending, patient-type (IN/OUT), consulting charge, Blood test charge, Ex-ray charge, other test charges and total fee. Write a report program for the following conditions :

(i) The patients, undergone blood test

(ii) The patients, who taken x-rays

(iii) the patients, who belong to in-patient category.

The common fields to be included in the above mentioned reports are : patient no, patient name, age and corresponding charge and total fee.

(b) Write a shell program to prepare electric bill for domestic consumers.

For first 100 units – Rs. 0.75/unit

For next 100 units – Rs. 1.50/unit

Above 200 units – Rs. 3.00/unit

Prepare the bill for the following format

Customer No. ————————

Customer Name ————————

Pre. reading ————————

Cur. reading ————————

Units consumed ————————

Charge ————————

8. (a) Write a SQL program for the canteen information system having two tables namely MENU and BILL tables. The MENU table having the following fields ITEM and UNIT-RATE.

The assumption here is that only the following items avail at the canteen : TEA, COFFEE and COOL DRINKS. The BILL table having EMPNO, EMPNAME, DATE OF ISSU, ITEM 1, NO. OF TOKENS FOR ITEM 1, RATE 1, ITEM 2, NO.OF TOKENS FOR ITEM 2, RATE 2, ITEM 3, NO. OF TOKENS FOR ITEM 3, RATE 3, TOTAL AMOUNT.

Note : An employee is allowed to get the token only once in a month. Where RATE is UNIT-RATE * NO. OF TOKENS for particular item and total amount is cumilation of rates.

Prepare the following report in the prescribed format as given below :

S.NO	EMPNO	EMPNAME	PURCHASED RATE
1	–	–	–
2	–	–	–
3	–	–	–
			Total amount

(b) Write a shell program to convert all capital letters in a file to small case letters.

DE–6630 **213**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

LAB IV — INTERNET AND VISUAL PROGRAMMING

(2003 and 2004 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Examiner has to choose ONE question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a java script to create a window by using the confirm message.

   (b) Write a VB program to perform the following operation in record of random access file :

   (i) Insert

   (ii) Delete

   (iii) Search

---------- Cut here ----------

2. (a) Write a java script to create a color palet and display the background in the color chosen from the palet.

   (b) Using built in AX control, develop the windows notepad with file and edit menu operation and also display the floating menu whenever necessary.

3. (a) Write a VB script to do the following :

   (i) Check the given password

   (ii) Change the existing password.

   (b) Create a worksheet with the fields student name, internal mark, external mark and result. Using data access object, create a student application to do insert, delete and update the records.

---------- Cut here ----------

4. (a) Using at least 20 HTML tags, create a screen with a string "WEB DESIGN".

   (b) Using AX DLL or EXE add a class module that would perform the following function :

   (i) Text weather the given number is perfect or not.

   (ii) Find the factorial of the given numbers.

---------- Cut here ----------

5. (a) By using database connectivity display, insert and delete the records from the table.

   (b) Write a visual C++ program to get the status of the shift and toggle keys using MFC.

---------- Cut here ----------

6. (a) Develop a web page for job recruitment agency in an IT industry.

   (b) Write a visual C++ program to fill background of the client area with a bitmap.

7. (a) Write a java program using applet to display the dialog and menu.

   (b) Write a visual C++ program to handle windows messages in MFC program.

---------- Cut here ----------

8. (a) Design and publish a web page for a college.

   (b) Write a visual C++ program to create a window of desired size using MFC.

DE–6631 **306**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2013.

LAB V — MULTIMEDIA SYSTEMS AND COMPILER DESIGN

(2003 and 2004 batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Examiner has to choose the question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C++ program to find the best grey level for splitting an image using automatic selection for splitting technique.

   (b) Write a C program to translate the expression $(a+b)*(b-c)/(a+b)$ into post fix notation, three address code, quadruples and triples.

---------- Cut here ----------

2. (a) Write a C++ program for the following grey level transformation techniques
   (i) Thresholding
   (ii) Bunching

   (b) Write a C program to construct an SLR parsing table for the following grammar

   $E \rightarrow E+T/T$

   $T \rightarrow T*F/F$

   $F \rightarrow (E)/id$

3. (a) Write a C++ program to transform the gray level using histogram equalization technique.

(b) Write a C program to construct a canonical LR parsing table for the following grammar.

$S \rightarrow CC$
$C \rightarrow cC / d$

---------- Cut here ----------

4. (a) Write a C++ program to convolute an image with the following template

1 0
0 1

(b) Write a C program to recognize identifier in the given source code.

---------- Cut here ----------

5. (a) Write a C++ program to compress the image using run-length encoding technique.

(b) Write a C program to recognize constant in the given source code.

---------- Cut here ----------

6. (a) Write a C++ program for 2 dimensional image animation using computer graphics.

(b) Write a C program which accepts $(a/b)^* abb$ and draw the transition diagram for DFA.

7. (a) Design a sales advertisement for a product using Flash.

   (b) Write a C program to convent infix to post fix notation.

---------- Cut here ----------

8. (a) Design an advertisement for a product using power point presentation.

   (b) Write a C program to accept a string and to check whether its satisfies the following production rule using shift reduce parsing.

$$E \rightarrow E + E \mid E - E \mid E * E \mid E / E \mid id \mid const \mid (E)$$